EE-170L Computer Programming Lab

Laboratory Report — No. 04

Switch Statement, break and continue in C++

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1. Lab Objectives

By the end of this lab session, students will be able to:

- Understand and use the switch statement as a multi-way selection tool in C++.
- Apply the break statement to terminate case blocks and prevent fall-through.
- Understand the role of the default section in handling unmatched cases.
- Use the continue statement inside loops to skip iterations.
- Differentiate clearly between break and continue and when to use each.
- Develop practical menu-driven programs using switch.

2. Theoretical Background

2.1 The switch Statement

The switch statement is used to select one block of code from multiple options based on the value of a single variable or expression. It is commonly used as a cleaner alternative to a long if-else-if chain when comparing one variable against several constant values.

Key characteristics of the switch statement:

- The switch expression must evaluate to an integer or character constant.
- Each case label must have a unique constant value — duplicate cases are not allowed.
- It falls under the category of selection statements, alongside if and else.
- It is called a multiple-selection statement because it can branch to many different actions.

Syntax:

```
switch (expression)
{
```

```
    case value1:
        // code block 1
        break;

    case value2:
        // code block 2
        break;

    default:
        // default code block (optional)
}
```

2.2 The break Statement

The break statement is used inside a switch block to terminate a case and prevent fall-through. Without break, execution continues into the next case regardless of whether it matches.

- Case statements mark where execution begins inside the switch block.
- break statements mark where execution ends for that case.
- Without break, all remaining cases execute until the closing brace or a break is found.
- break is optional — intentional fall-through is valid C++ but should be commented clearly.

2.3 The default Section

The default section is an optional part of a switch that executes when no case value matches the switch expression. It acts as a catch-all and is typically placed at the end of the switch block.

- The default section is not required; the program will run fine without it.
- If included, it handles all unmatched inputs and prevents silent failures.

2.4 The continue Statement

The continue statement is used inside loops (for, while, do-while) to skip the rest of the current iteration and jump directly to the next iteration. Unlike break, it does not exit the loop entirely.

- **break:** Exits the loop or switch block immediately.
- **continue:** Skips the current loop iteration and moves to the next one.

3. Example Program — Simple Calculator Using switch

The following program demonstrates how switch is used to build a simple four-operation calculator. The operator character entered by the user determines which arithmetic operation is performed:

```
#include <iostream>
using namespace std;

int main()
{
    int  num1, num2;
    char op;

    cout << "Enter two numbers: ";
    cin  >> num1 >> num2;
    cout << "Enter operator (+, -, *, /): ";
```

```
    cin  >> op;

    switch (op)
    {
        case '+':
            cout << "Result = " << num1 + num2;
            break;

        case '-':
            cout << "Result = " << num1 - num2;
            break;

        case '*':
            cout << "Result = " << num1 * num2;
            break;

        case '/':
            cout << "Result = " << num1 / num2;
            break;

        default:
            cout << "Invalid operator!";
    }
    return 0;
}
```

4. break vs continue — Key Differences

Feature	break	continue
Purpose	Exits the loop or switch block	Skips current iteration of the loop
Used in	Loops and switch statements	Loops only (for, while, do-while)
Effect	Terminates the entire loop/switch	Moves directly to the next iteration
Remaining code	Skipped; exits the block immediately	Skipped for current iteration only

Example — continue skipping odd numbers:

```
#include <iostream>
using namespace std;

int main() {
    // Print only even numbers from 1 to 10
    for (int i = 1; i <= 10; i++) {
        if (i % 2 != 0)
            continue;   // skip odd numbers
        cout << i << " ";
    }
    return 0;
}

// Output: 2 4 6 8 10
```

5. Conclusion

This lab provided a practical understanding of program flow control in C++ using the switch, break, and continue statements. The key takeaways are:

- **switch Statement:** A clean and efficient way to handle multi-way branching based on a single variable's value, replacing lengthy if-else-if chains.
- **break Statement:** Essential within switch cases to prevent unintended fall-through; also used to exit loops early when a condition is met.
- **default Section:** An optional but recommended safety net in switch that catches all unmatched case values.
- **continue Statement:** Used inside loops to skip the remainder of the current iteration and proceed directly to the next, without exiting the loop.
- **Menu-Driven Programs:** The switch statement is ideal for building interactive menus where each option maps to a distinct code block.

Understanding these control flow tools is fundamental for writing clear, efficient, and correct C++ programs. They will be applied extensively in upcoming labs involving loops and functions.